

Binary consequences of relation-liveness automata lower bounds

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Abstract

We give explicit fixed-binary-alphabet consequences of OpenAI’s two relation-liveness lower bounds. Encoding each relation on h points by its h^2 adjacency bits gives a language recognized by a one-way nondeterministic automaton with $N_h = (3h^3 - h)/2 + 2$ states. Every equivalent two-way deterministic automaton, and every two-way nondeterministic automaton for its full binary complement, requires $2^{\Omega(h)} = 2^{\Omega(N_h^{1/3})}$ states. We specify all transition rules, including an exact sh^2 -state pullback for an s -state binary target. The binary language rejects incomplete blocks; a fooling set proves that its one-way nondeterministic state complexity is $\Theta(h^3)$. The exponential obstructions are inherited from the two upstream theorems; binary adjacency coding is established machinery.

1 Scope, provenance, and conventions

Sakoda and Sipser introduced the state cost of deterministic simulation of two-way nondeterministic automata [1]. Here we take as inputs the two explicit relation-language lower bounds in [2, 3]. Both sources state their growing-alphabet scope. Our contribution is an explicit transfer to one binary alphabet with full-word semantics, exact state counts, and a matching cubic order for the chosen source encoding. We do not give an independent proof of the upstream exponential obstructions.

Adjacency-bit encoding already appears in Kapoutsis’s discussion of binary one-way liveness [4, Section 5] and [5, Section 8], there for deterministic targets with few reversals. That discussion does not specify the treatment of incomplete binary blocks. Our cubic source-size claim concerns exactly the total language defined below. Other encodings can have different source costs. A public binary-construction record [9] also uses standard coding and target pullback, with a quadratic inherited lower bound. Neither antecedent supplies the unrestricted exponential inputs used here. We make no claim to have invented binary coding or to have independently resolved the base conjecture.

For comparison, Adeogun–Kapoutsis [6, Theorem 1] proved a quadratic unrestricted deterministic lower bound for relation liveness. Their later method limitation [7, Theorem 2.1] bounds a particular chain method, rather than every possible proof. Guillon–Prigioniero–Taheri [8, Section 1] describe read-only 2NFA complementation as open; their polynomial complementation theorem uses 1-limited automata, which may rewrite cells. Our results concern read-only machines. The upstream manuscript dates are September 25, 2026; the public repository release was verified on October 6, 2026. These dates should not be conflated with priority claims.

Our marked model has one initial state and a set of accepting states. The head starts on the left endmarker of $\langle x \rangle$; the two endmarkers are distinct, including on empty input. A transition depends on the state and scanned symbol, chooses a new state and a move in $\{-1, 0, 1\}$, and moves

the head accordingly. Outward moves at endmarkers are disabled. Partial transitions are allowed. Acceptance is reachability of an accepting state by a finite run, including zero transitions. Missing transitions and infinite nonaccepting runs do not accept. A 2DFA has at most one transition per row; a 2NFA may have several. All states are counted. Our one-way source moves right on every body bit; its only stay is the accepting move at the right marker. We also give an ordinary epsilon-free 1NFA, whose acceptance is tested after consuming the whole word. The distinction matters for empty input. Requiring a positive finite accepting run is also covered below.

2 Language and exact source compiler

Fix $h \geq 2$, let $H = \{0, \dots, h-1\}$, and let Σ_h be all relations on H . Products are in path order: $(p, r) \in RS$ exactly when some q has $(p, q) \in R$ and $(q, r) \in S$. Write

$$L_h = \{R_1 \cdots R_k \in \Sigma_h^* : R_1 \cdots R_k \neq \emptyset\}, \quad R_1 \cdots R_0 = I_H.$$

Set $b = h^2$. The bit at offset $t = ph + q$ of $\text{enc}(R)$ is 1 exactly when $(p, q) \in R$. Extend enc by concatenation, with $\text{enc}(\varepsilon) = \varepsilon$, and define on *every* binary word

$$K_h = \text{enc}(L_h) \subseteq \{0, 1\}^*.$$

Every length- b binary block codes exactly one relation. Thus K_h rejects all words whose lengths are not divisible by b , and otherwise tests nonemptiness of the decoded product. It contains ε .

Lemma 1. *K_h has a marked one-way NFA with exactly $N_h = (3h^3 - h)/2 + 2$ states, and an ordinary epsilon-free 1NFA with $N_h - 1$ states.*

Proof. The marked states are a nonaccepting initial state i , an accepting state a , and

$$U(p, t) \quad (0 \leq p < h, \ 0 \leq t < (p+1)h), \quad V(q, t) \quad (0 \leq q < h, \ q+1 \leq t < b).$$

The offset is the next bit to be read. U retains the preceding vertex before choosing an outgoing edge; V retains the chosen next vertex. The rules at the left marker are $i \xrightarrow{\triangleleft, +1} U(p, 0)$ for all $p \in H$. At the right marker, precisely the states $U(p, 0)$ have the rule $U(p, 0) \xrightarrow{\triangleright, 0} a$. State a has no rules. For a body bit c , give $U(p, t)$ the following rules:

- If $t < (p+1)h - 1$, it may skip the bit and move right to $U(p, t+1)$.
- If $c = 1$ and $\lfloor t/h \rfloor = p$, put $q = t \bmod h$. It may move right to $V(q, t+1)$ when $t < b-1$, and to $U(q, 0)$ when $t = b-1$.

From $V(q, t)$ either body bit moves right to $V(q, t+1)$ for $t < b-1$, and to $U(q, 0)$ for $t = b-1$. All rules not listed are absent. Every listed destination is in the displayed state ranges: a chosen edge at offset $ph + q$ gives $t+1 \geq q+1$. There are no left moves.

A surviving branch through a full block must choose exactly one 1 in the row of its preceding vertex. It then arrives at the next block as $U(q, 0)$, precisely when the block relation contains that edge. Skipping the entire row is impossible because the last row offset has no skip rule. Induction on complete blocks identifies surviving boundary vertices with endpoints of paths through the decoded relations. Only complete boundaries can accept at the right marker. This proves rejection of partial blocks as well as correctness on valid words. On empty input, the initial right move and the accepting stay give acceptance because H is nonempty. All accepting runs have at least two transitions, so positive-run acceptance gives the same source language.

The two families have sizes

$$\sum_{p=0}^{h-1} (p+1)h = \frac{h^3 + h^2}{2}, \quad \sum_{q=0}^{h-1} (b-q-1) = h^3 - \frac{h^2 + h}{2}.$$

Adding i, a proves the marked count. For an ordinary 1NFA, retain U, V , replace i, a by one new initial state i_0 , and make i_0 and all $U(p, 0)$ final. Its bit row at i_0 is the union of the bit rows at $U(p, 0)$ over all p . All other bit rows are as above, with their right moves interpreted as ordinary consumption. There are no transitions to i_0 . The empty word accepts at i_0 ; every nonempty word simulates the first bit from some starting vertex and then the marked source's bit computation. Final states test full consumption at a complete boundary. The count is $N_h - 1$. $\square$

Proposition 2. *Every ordinary 1NFA recognizing K_h has at least h^3 states. The same bound holds for a marked right-only source in our finite-run model. Consequently the one-way state complexity of this strict-block language is $\Theta(h^3)$.*

Proof. Let $P_p = \{(p, p)\}$ and $z = \text{enc}(I_H)$. For $p \in H$ and $0 \leq t < b$, put

$$x_{p,t} = \text{enc}(P_p) \text{pref}_t(z), \quad y_{p,t} = \text{suff}_{b-t}(z) \text{enc}(P_p).$$

Here the prefix and suffix lengths add to b , including $t = 0$. For each matching pair, $x_{p,t}y_{p,t}$ encodes $P_p I_H P_p = P_p$, so belongs to K_h . A crossed pair with $t \neq u$ has length $3b + t - u$, which is not divisible by b , since $0 < |t - u| < b$. When $t = u$ but $p \neq r$, the crossed word encodes $P_p I_H P_r = \emptyset$. Thus all $hb = h^3$ diagonal pairs accept and all off-diagonal crossed pairs reject.

Choose an accepting run for each diagonal pair and its state at the cut between x and y . If two cuts had the same state, splicing the prefix of one run to the suffix of the other would accept a crossed word. The cuts must therefore occupy distinct states. This is the standard fooling-set argument, here with its explicit witnesses. For a marked source, an accepting run cannot accept before visiting the right marker: otherwise the same finite prefix of its run accepts an extension by one bit, whose length is malformed. Such a run therefore has a cut after the prefix before reading the suffix, and the same splicing argument applies. Stay moves, if allowed in a right-only source, can be assigned to either side of the cut and do not affect the argument. $\square$

3 Exact pullback for all two-way targets

Lemma 3. *An s -state binary 2NFA T has an explicit sb -state 2NFA T^{enc} over Σ_h satisfying*

$$T^{\text{enc}} \text{ accepts } w \iff T \text{ accepts } \text{enc}(w).$$

Determinism is preserved. This equivalence holds for both zero-step and positive-run acceptance, with partial transitions and nonaccepting infinite runs allowed.

Proof. If Q is the state set of T , use $Q \times \{0, \dots, b-1\}$. The initial state is $(q_0, 0)$; (q, t) accepts exactly when q accepts. At a relation letter R , inspect its bit at offset t . For every transition $(q, c) \mapsto (q', d)$ of T , where $c = \text{enc}(R)_t$, use

d	new offset	real move
0	t	0
$+1, t < b-1$	$t+1$	0
$+1, t = b-1$	0	$+1$
$-1, t > 0$	$t-1$	0
$-1, t = 0$	$b-1$	-1

with first state coordinate q' . At the left marker, use T 's left-marker row independently of t : a stay retains t , and a right move sets it to 0 and moves right. At the right marker, use its right-marker row: a stay retains t , and a left move sets it to $b - 1$ and moves left. Outward moves are disabled as in T . No transition is supplied for a missing row.

On a relation word of length k , map a configuration at real body cell $j \in \{1, \dots, k\}$ with state (q, t) to the binary configuration with state q at position $(j - 1)b + t + 1$. At either real marker, map all phases to the corresponding binary marker with state q . The rules above show that each transition maps to exactly one transition of T , and that every transition of T from the image configuration lifts from *each* corresponding pullback configuration. Marker phases need not be unique, because marker rows ignore them. On the empty word, moves between its adjacent markers remain correct under this mapping. Starting from the prescribed initial states, induction on run length proves finite-run equivalence, with the same length. Acceptance and time-zero acceptance are preserved; hence positive-run acceptance is preserved too. Transition-by-transition lifting also preserves infinite runs. An infinite run supplies no new finite accepting witness on either side. The construction has exactly sb states and at most one rule per row if T does. $\square$

4 The two binary lower bounds

We state explicitly the two external results used. For $h \geq 2$, every s' -state marked 2DFA for L_h satisfies

$$2^{\lfloor (h-2)/31 \rfloor} \leq 4(s' + 2)^2 \quad (1)$$

under either finite-run convention [2, Theorem 1.1]. Under zero-step acceptance its stronger bound uses $s' + 1$. Every s' -state marked 2NFA for $\Sigma_h^* \setminus L_h$ satisfies

$$2^{\lfloor (h-2)/127 \rfloor} \leq 2(s' + 1) \quad (2)$$

[3, Theorem 1.1]. The latter source uses zero-step acceptance. Both inputs quantify over all finite relation words and unrestricted read-only two-way targets.

Theorem 4. *For every $h \geq 2$, the explicit language $K_h \subseteq \{0, 1\}^*$ of Section 2 has the source bounds of Lemma 1. If an s -state marked 2DFA recognizes K_h , then*

$$2^{\lfloor (h-2)/31 \rfloor} \leq 4(sh^2 + 2)^2, \quad s \geq \frac{2^{\lfloor (h-2)/31 \rfloor / 2 - 1} - 2}{h^2}. \quad (3)$$

If an s -state marked 2NFA recognizes its complement in all binary words, then

$$2^{\lfloor (h-2)/127 \rfloor} \leq 2(sh^2 + 1), \quad s \geq \frac{2^{\lfloor (h-2)/127 \rfloor - 1} - 1}{h^2}. \quad (4)$$

In particular, both target state costs are $2^{\Omega(N_h^{1/3})}$. There is no polynomial state bound for either simulation or complementation on this one fixed alphabet.

Proof. Fixed-length coding is injective on all Σ_h^* and $\text{enc}(w) \in K_h$ if and only if $w \in L_h$. Apply Lemma 3 to a deterministic binary recognizer and insert $s' = sh^2$ into (1). Taking square roots gives (3). For complementation, the full universe satisfies

$$\{0, 1\}^* \setminus K_h = \{x : b \nmid |x|\} \cup \text{enc}(\Sigma_h^* \setminus L_h),$$

a disjoint union. The pullback sees valid codewords, and for *every* relation word w ,

$$\text{enc}(w) \in \{0, 1\}^* \setminus K_h \iff w \in \Sigma_h^* \setminus L_h.$$

Thus it recognizes the entire upstream complement. Apply (2) with $s' = sh^2$ and rearrange. Empty input is absent from both complements; malformed binary inputs are included in the binary complement. Finally $h^3 \leq N_h \leq 2h^3$ for $h \geq 2$. The exponential factors in (3)–(4) absorb polynomial denominators and constant subtractions as $h \rightarrow \infty$, proving the stated asymptotics. The ordinary source count $N_h - 1$ has the same cubic order. $\square$

For a complementing target using positive-run acceptance, add a new nonaccepting initial state with the old initial state’s transition row, whose destinations remain old states. Under zero-step acceptance this $s+1$ -state machine accepts exactly the old positive-run language: its first transition is the required positive first step. Hence the complementation statement remains stretched exponential, with the explicit variant $2^{\lfloor (h-2)/127 \rfloor} \leq 2((s+1)h^2 + 1)$. The deterministic formula already covers both conventions. If a convention requires total transition rows, an additional nonaccepting sink with legal stay loops supplies missing rules; adding a sink changes the source bound by one and does not weaken a lower bound for the larger partial model. Conventions requiring acceptance at the right endmarker or halting before accepting form subclasses of the target model; the same lower bounds apply.

The exponent here is proportional to the cube root of the binary source size. We do not assert $2^{\Omega(n)}$ for n source states, or a uniform complexity separation such as $L \neq NL$. The transition tables of relation-alphabet pullbacks can be large; this is a state-count argument, not a small-table uniform algorithm claim.

5 Verification and disclosure

The upstream pin is `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Independent automated audits inspected both original proof mechanisms, the actual Lean theorem statements and machine semantics, and finite algebraic counterexample checks. The new reduction is proved above; its compiler and exhaustive tiny-instance simulator accompany the paper. The simulator tests malformed lengths, empty input, legal marker behavior, partial transitions, stays, both head directions, initial acceptance, positive acceptance, and loops by configuration reachability. Finite tests are falsification checks, not substitutes for uniform proofs. A project-local attempt to rebuild the upstream Lean inputs was prevented by insufficient disk capacity. We make no claim of reproduced formal verification, and the binary results in this note are not formalized. The package records exact source hashes, commands, scope and limitations.

AI tools were used extensively in research, drafting and verification. Independent automated adversarial reviews are documented with the package. This preprint has not undergone conventional human peer review or refereeing. The upstream model-produced work is attributed to OpenAI as instructed by its manuscript-specific citations. No affiliation or coauthor is asserted.

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